

Final Report - Team 4.1 - 4.2



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Entrepreneurship

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Our Take on Responsible Entrepreneurship and Why It Matters

Entrepreneurship, at its essence, is about creating something new through spotting an opportunity, acting on it, and building value where none existed before. Entrepreneurs take risks, solve problems, and push ideas into reality, often shaping the way industries and societies evolve. Whether the goal is financial success, creative expression, or social impact, entrepreneurship thrives on initiative, resilience, and a willingness to challenge the norm. But in today's world, where the consequences of business decisions reach far beyond profit margins, a new kind of entrepreneurship is emerging, one that goes beyond building a viable product or capturing market share. This is where *responsible entrepreneurship* comes in.

Responsible entrepreneurship is about building companies that not only adapt, but also consciously contribute to society and continuously improve how they operate. It's about extending the company's vision beyond having a great product or securing the most investment by doing business in a way that aligns with long-term societal wellbeing, environmental stewardship, and ethical responsibility whilst smartly keeping sight of finances. The main difference we believe to exist between regular entrepreneurship and *responsible* entrepreneurship is a shift in the goal of the company's existence to not just maximize value for shareholders but do so in a way that doesn't compromise on the benefits of other stakeholders such as employees, customers, communities, and the environment. In this entrepreneurial model, success isn't measured solely by growth or market dominance, but by how well the business balances its impact with its income. Responsible entrepreneurs ask: *Who is affected by our decisions? What are the long-term consequences? Are we building not products but also practices that genuinely improves lives for all groups of society?*

This shift reflects a growing understanding that companies don't operate in a vacuum. They're part of larger systems, economic, social, ecological, and they have a role to play in maintaining the health of those systems. It's not about being perfect, but about being intentional. Responsible entrepreneurs acknowledge trade-offs and strive to make better choices with the information and resources available. They're transparent with stakeholders, willing to listen and adapt, and they embed responsibility into their core business logic, not as a marketing strategy, but as a guiding principle.

This kind of entrepreneurship matters because it changes what success looks like. It's no longer just about market share or valuation. It's about creating a measurable, lasting impact. Ultimately, responsible entrepreneurship matters because the world is changing. Crises like climate change, inequality, and public health demand more from business leaders. Entrepreneurship must evolve from being simply opportunity-driven to being *responsibility-driven*. For Azure Cell Therapies, that extends beyond manufacturing advanced therapies to setting concrete ethical practices and integrating them throughout all the different stages of the product's life cycle as **responsible entrepreneurship is what makes innovation not just impressive, but also meaningful.**

Executive Summary Week 3

During session 3, the focus was on defining Azure Cell Therapies' value proposition and articulating its core identity through purpose, vision, and mission statements. This step was key to aligning the company's innovation with stakeholder needs and long-term societal impact, especially given its goal to offer a one-time stem cell therapy for Parkinson's disease.

The session began by mapping out Azure's key target audiences: hospital managers (customers), doctors (users), and Parkinson's patients (beneficiaries). Using a human-centered approach, we analyzed their specific pain points. Ranging from limited treatment options to challenges in affordability and long-term effectiveness. Azure's solution, a disease-modifying therapy using neuron replacement, was reframed to highlight both its medical potential and its broader social value.

We then crafted targeted value proposition statements. Rather than focusing solely on technical performance, we integrated sustainability and patient-centered outcomes, learning how to communicate innovation in a way that resonates with both professionals and those directly affected by the disease.

In the second part, we developed purpose, vision, and mission statements that reflected Azure's ambition to reshape neurodegenerative disease treatment responsibly. The purpose focused on addressing a critical unmet need; the vision imagined a world where Parkinson's no longer inspires fear; and the mission emphasized safe, scalable, and lasting therapies grounded in ethics and science.

Overall, this session helped us understand how a powerful biotech narrative is built, not just on scientific achievement, but on a clear, empathetic, and sustainable promise to society.

Group 4.1 	Group 4.2 
	

Executive Summary Week 4

During session 4, the session focused on understanding and measuring the broader impact of Azure Cell Therapies beyond financial success. Using the Theory of Change framework, we explored how the company's activities, such as research, development, and delivery of its stem cell therapy, can lead to meaningful outcomes for patients, healthcare systems, and society.

We began by identifying the most relevant Sustainable Development Goals, particularly SDG 3 (Good Health and Well-Being) and SDG 9 (Industry, Innovation and Infrastructure). We then discussed how Azure could contribute to these goals by improving access to safe and effective treatments while driving innovation in biotech.

The core of the session involved building a Theory of Change. We mapped the company's key inputs, activities, and outputs, and linked them to short- and long-term outcomes. For each output, we identified the stakeholders affected, including patients, doctors, and hospitals, and defined how they benefit. We also proposed indicators that could help Azure track its impact over time.

Finally, we addressed common challenges in impact measurement, such as proving causality, managing long time horizons, and avoiding greenwashing. Despite these difficulties, we saw how early thinking around impact can help startups grow more responsibly and with greater clarity.

This session reinforced the idea that strong innovation in healthcare must be grounded in measurable, long-term value for both people and the planet.

Group 4.1 	Group 4.2 
	

Executive Summary Week 5

In Week 5, both Groups 4.1 and 4.2 focused on evaluating the environmental impact of Azure Cell Therapies' core product: a single dose of stem cell therapy for Parkinson's disease. Using the Life Cycle Assessment (LCA) method, we mapped out the full journey of this product: from drug development all the way to waste disposal. For each stage, we identified the key materials, energy inputs, and types of waste generated, and then linked them to their environmental impacts, such as carbon emissions, resource use, and toxicity.

Through our research, we found that the drug development phase is where the biggest environmental burden lies, mainly due to the large amount of single-use plastic consumed in the lab, which is then incinerated. This alone contributes around 676.8 kg of CO₂ emissions each month, and that number could grow as the company scales up. Since this stage is still fully under the start-up's control, we saw a clear opportunity for improvement.

To address this, we suggested practical steps like tracking plastic use more closely, switching to reusable labware where possible, and even investing in environmental offset programs. We also recommended aiming for a My Green Lab certification to show a real commitment to sustainability. Overall, this session helped us think critically about how a biotech start-up can grow responsibly, by recognizing its environmental impact early and taking meaningful steps to reduce it.

Group 4.1 	Group 4.2 
	

Executive Summary Week 6

In Week 6, we took a closer look at the potential risks and long-term sustainability of Azure Cell Therapies' innovative treatment, a one-time stem cell therapy designed to help patients with Parkinson's disease. Building on the work from previous weeks, our goal was to understand not just the positive outcomes of the therapy, but also the unintended negative consequences it could bring, especially as the company starts to grow.

We began by doing a Theory of Change – Risk Analysis, identifying four key areas of concern. One major issue was the possibility that patients or doctors might misunderstand the therapy's benefits, expecting a permanent cure when the actual outcome is "sustained improvement." This could lead to legal issues or harm the company's reputation. Another concern involved the risk of side effects, especially neurological or immune complications, which might affect certain patient groups more than others. We also looked at production risks: scaling up stem cell manufacturing could lead to inconsistencies, with some batches being less effective or even unsafe. Finally, we considered the danger of the technology being misused by unauthorized clinics or researchers, which could damage public trust in the therapy.

In the second part of our work, we pulled together everything we've learned so far, from our initial Theory of Change to our Life Cycle Assessment (LCA) and the new risk analysis, into a set of actionable recommendations. The LCA reminded us that the biggest environmental impact right now comes from plastic waste during drug development, which leads to significant CO₂ emissions. We suggested using more sustainable lab materials, tracking plastic usage more carefully, and investing in carbon or plastic offset programs.

Overall, this week helped us think more holistically about Azure Cell Therapies, not just as a promising biotech start-up, but as a company that has the chance to grow responsibly, with a strong awareness of its social and environmental footprint.

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Executive Summary Week 7

During session 7, groups worked on developing a comprehensive and sustainability-oriented Business Model Canvas (BMC) for Azure Cell Therapies. The aim was to systematically assess how the startup could create, deliver, and capture value while aligning its strategy with ethical and environmental sustainability principles. Each building block of the BMC was examined in detail, including key customer segments such as neurology and movement disorder clinics, hospitals with neurodegenerative programs, and big pharmaceutical companies, with a focus on their demographics, needs, and access challenges.

The group defined Azure Cell Therapies' value proposition around delivering a one-time, disease-modifying neuron replacement therapy for Parkinson's disease. Key activities highlighted included R&D, strategic partnerships, scalable manufacturing, regulatory compliance, and essential search for entrepreneurial support like funding, mentoring and business development. Partnerships with logistics providers, manufacturers, and payers were identified as critical to enabling market access and distribution. The importance of transparent, evidence-based communication, long-term customer relationships, and specialized ongoing support and training was also emphasized.

Sustainability questions were integrated across the canvas, exploring how the use of green chemistry principles, reduction of single-use plastics, and optimization of cold-chain logistics could mitigate ecological costs while enhancing cost-efficiency and investor appeal. Groups emphasized the fact that although upfront costs are high, embedding sustainability into the company's business model can strengthen long-term value, improve regulatory alignment, and support broader social impact in the healthcare system.

Group 4.1 	Group 4.2 
	

Executive Summary Week 9

During session 9, groups conducted a comprehensive STEEPLE analysis to evaluate the external factors that influence the strategic direction, competitiveness, and sustainability of Azure Cell Therapies. The objective was to assess key Social, Technological, Economic, Environmental, Political, Legal, and Ethical trends and their potential to either create opportunities or pose risks to the startup.

Using AI-generated insights as a starting point, groups identified and critically reflected on the most relevant trends. Key social insights included the growing demand for Parkinson's treatments due to aging populations and increased awareness of neurodegenerative conditions, as well as equity concerns related to access. Technological trends, such as advances in stem cell practices, AI in manufacturing, and the use of digital platforms to monitor patient outcomes, were recognized as both enablers and competitive threats. Economically, students highlighted high R&D costs, regional funding disparities, and the impact of public versus private insurance systems on therapy adoption. Environmentally, the analysis stressed the need for sustainable production, carbon-conscious logistics, and responsible sourcing. Political and legal reflections covered global healthcare policy alignment, clinical trial regulations, and IP protection, while ethical considerations centered on inclusivity, transparency, and equitable access.

Moreover, groups conducted a competitor analysis, comparing Azure Cell Therapies to leading biotech firms specializing in treatment for PD using dimensions such as safety, scalability, patents, and long-term therapeutic potential. This informed a refined value proposition, as well as clearer mission and vision statements, reinforcing the company's commitment to innovation, accessibility, and sustainability. Overall, this session emphasized the importance of embedding resilience and ethics into strategic planning to ensure Azure's relevance in a fast-evolving, impact-driven healthcare landscape.

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Executive Summary Week 10

During session 10, groups focused on identifying and evaluating alternative market opportunities for Azure Cell Therapies using the Market Opportunity Navigator framework. The objective was to explore how the startup's core assets such as expertise in cell therapy, advanced neuron manufacturing technologies, and access to funding could be leveraged beyond the company's primary offering for Parkinson's disease treatment.

Groups generated a diverse set of market opportunities by analyzing unmet needs in related sectors, guided by insights from previous STEEPLE analysis. Five promising opportunities were evaluated for their strategic and societal attractiveness. These included biobanking services as a strategic opportunity, consultancy services and collaboration with PD students, neurotoxicity testing for cosmetics and pharma, stem cell therapy for injuries/traumas/strokes, and neurotech-powered art installations and immersive experiences. Each was assessed based on external risks, feasibility, market potential, competitive advantage, and positive societal impact.

Finally, an agile focus strategy was designed, highlighting Azure Cell Therapies' primary target market while identifying attractive secondary opportunities for future expansion. This exercise reinforced the importance of strategic flexibility and demonstrated how sustainability and innovation can open new commercial avenues, helping the company to create broader societal value while building a resilient business model.

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Azure Cell Therapies: Strategic Analysis & Recommendations

Executive Summary

Azure Cell Therapies has positioned itself as a pioneering biotech startup addressing Parkinson's disease through advanced, scalable, tumor-safe stem cell therapies aimed at sustained neuron regeneration. The company's proprietary technology platform differentiates itself in a competitive landscape by tackling critical issues of scalability, tumor safety, and sustained efficacy that have historically limited cell therapy approaches for neurological conditions. Through its innovative approach, Azure aims to transform treatment paradigms while establishing sustainable market leadership in the rapidly evolving cell therapy sector.

Market Position & Competitive Advantages

The company's core differentiation lies in its patented technology that enhances scalability, ensures tumor safety, and maintains long-term effectiveness. Compared to direct competitors such as BlueRock, Sumitomo, and Novo Nordisk, Azure uniquely offers comprehensive solutions across critical dimensions including scalable manufacturing, tumor safety, and sustained efficacy, although it remains at the preclinical stage.

The competitive landscape reveals that while other companies focus on individual aspects of cell therapy development, Azure's integrated approach addresses multiple therapy limitations simultaneously. This comprehensive strategy positions the company to capture significant market share as the technology advances through clinical development phases and toward commercialization.

Strategic Recommendations

Regulatory Pathway Acceleration: Azure should establish dedicated regulatory affairs capabilities and initiate pre-IND meetings with regulatory bodies to streamline the path to clinical trials. Given the competitive landscape with established players advancing their programs, early engagement with regulators will be critical to maintaining Azure's technological advantages through faster market entry and efficient regulatory navigation.

Patient-Centric Clinical Design: The company should prioritize developing innovative clinical trial methodologies that reduce patient burden while generating robust efficacy data. This includes exploring decentralized trial elements, digital biomarkers for Parkinson's progression monitoring, and patient-reported outcome measures that better capture the sustained benefits of their therapy, ultimately improving patient engagement and data quality.

Intellectual Property Monetization: Azure should actively pursue licensing opportunities for its proprietary tumor-safety and scalability technologies to other cell therapy developers. This strategy will generate early revenue streams while validating the broader applicability of Azure's innovations beyond Parkinson's disease applications, creating multiple pathways to financial sustainability.

Healthcare System Integration: The company should establish pilot programs with leading academic medical centers and integrated health systems to generate real-world evidence and build key opinion leader support. These partnerships will provide valuable clinical insights while creating pathways for future commercial adoption and market penetration.

Risk Management & Sustainability Framework

Azure faces several potential risks including patient misunderstandings regarding sustained effects, neurological and immune complications, batch inconsistencies, and potential technology misuse. The company's mitigation strategies should involve clear patient communication programs, rigorous quality control systems, robust clinical trial designs with comprehensive safety monitoring, and stringent geo-locked treatment licenses to prevent unauthorized use.

The startup recognizes significant environmental impacts in drug development, particularly through plastic waste and CO₂ emissions from incineration processes. Azure should implement a comprehensive ESG framework that addresses the environmental footprint of biotech operations through circular economy principles in laboratory operations, transition to renewable energy sources for facilities, and establishment of measurable targets for waste reduction and carbon neutrality. Additionally, the company should develop a robust social impact measurement system that tracks patient access initiatives and community health outcomes, positioning Azure as a leader in responsible biotechnology innovation.

Health Economics & Market Access

Given the high-cost nature of cell therapies, Azure must proactively develop robust health economic models demonstrating long-term cost-effectiveness compared to current standard-of-care treatments. This includes establishing outcomes research capabilities and building relationships with health technology assessment bodies to ensure successful reimbursement and market access strategies. The company's approach should emphasize value-based care partnerships with payers and comprehensive reimbursement strategies that highlight the sustained benefits and reduced long-term healthcare costs associated with their innovative therapy.

Stakeholder Engagement & Revenue Diversification

Azure emphasizes strong partnerships with healthcare institutions, patient advocacy programs, and academic collaborations, maintaining commitment to transparency, ethical clinical practices, and inclusive patient access, especially for underserved communities. Strategic partnerships and diversified revenue streams, including technology licensing and early-access patient programs, can further bolster Azure's market presence while reducing dependency on single revenue sources and creating multiple pathways to commercial success.

The company's stakeholder engagement strategy should focus on building trust and credibility within the medical community while ensuring that patient voices remain central to development decisions. This approach will be critical for successful market penetration and long-term sustainability in the competitive biotechnology landscape.